

Computation of Magnetized Dynamic Friction Force Based on a Reduced Binary Interaction Model

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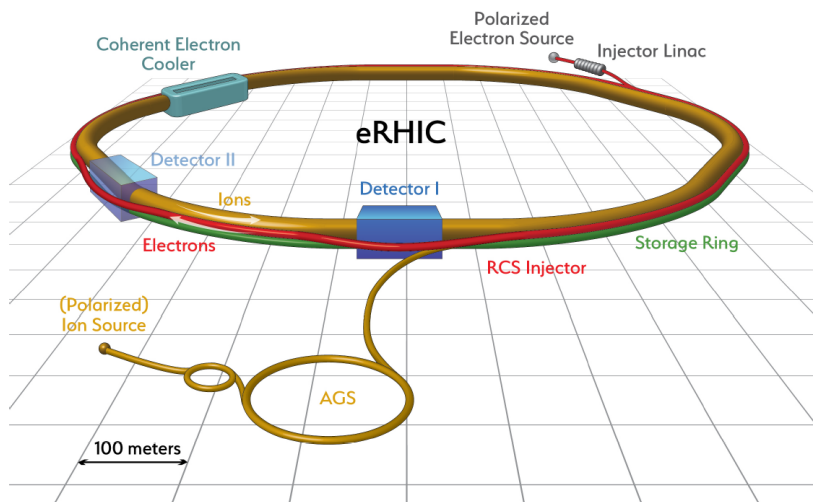
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Magnetized relativistic cooling is considered for the next-generation EIC designs

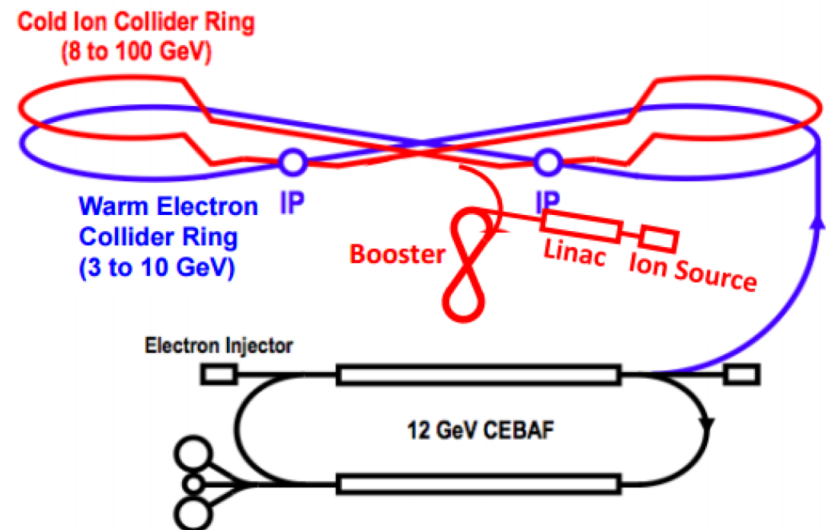
- Design of a polarized electron-ion collider (EIC) is a high priority for the nuclear physics community
- Relativistic, strongly-magnetized electron cooling
 - *may be essential for EIC, but never demonstrated experimentally*

eRHIC concept from BNL



C. Montag, “eRHIC Accelerator Design Overview,”
<https://www.jlab.org/indico/event/281/session/0/contribution/3/material/slides/0.pptx>

JLEIC concept from Jefferson Lab



S. Abeyrante et al., “MEIC Design Summary,”
<https://arxiv.org/pdf/1504.07961.pdf>

Relativistic cooling: short interaction time, new physics

- EIC requires cooling at high energy
 - $100 \text{ GeV}/n \rightarrow \gamma \approx 107 \rightarrow 55 \text{ MeV bunched electrons}, \sim 1 \text{ nC}$
- Electron cooling at $\gamma \sim 100$ requires different thinking
 - *friction force scales like $1/\gamma^2$ (Lorentz contraction, time dilation)*
 - challenging to achieve the required dynamical friction force
 - not all of the processes that reduce the friction force have been quantified in this regime $\rightarrow$ significant technical risk
 - *normalized interaction time is reduced to order unity*
 - $\tau = t\omega_{pe} \gg 1$ for nonrelativistic coolers
 - $\tau = t\omega_{pe} \sim 1$ (in the beam frame), for $\gamma \sim 100$
 - violates the assumptions of introductory beam & plasma textbooks
 - breaks the intuition developed for non-relativistic coolers
 - as a result, the problem requires careful analysis

Previous work: asymptotic model for cold, strongly magnetized electrons

$$F_{\parallel} = -2\pi Z^2 n_e m_e (r_e c^2)^2 \left[3 \left(\frac{V_{\perp}}{V_{ion}} \right)^2 \ln \left(\frac{\rho_{\max}^A}{\rho_{\min}^A} \right) + 1 \right] \frac{V_{\parallel}}{V_{ion}^3}$$

For large V_{ion} parallel to B :

$$F_{\parallel}(V_{\perp} = 0) = -2\pi Z^2 n_e m_e (r_e c^2)^2 \frac{1}{V_{\parallel}^2}$$

diverges as $V_{ion} \rightarrow 0$

$$r_L = V_{rms,e,\perp} / \Omega_L(B_{\parallel})$$

$$\rho_{\min}^A = \max(r_L, \rho_{\min})$$

$$\rho_{\max}^A = \min(r_{beam}, \rho_{\max})$$

$$\rho_{\max} = V_{rel} / \max(\omega_{pe}, 1/\tau)$$

$$V_{rel} = \max(V_{ion}, V_{e,rms,\parallel})$$

$$V_{ion}^2 = V_{\parallel}^2 + V_{\perp}^2$$

Ya. Derbenev, “Theory of Electron Cooling,” arXiv (2017); <https://arxiv.org/abs/1703.09735>

Ya. S. Derbenev and A.N. Skrinsky, “The Effect of an Accompanying Magnetic Field on Electron Cooling,” Part. Accel. **8** (1978), 235.

Ya. S. Derbenev and A.N. Skrinskii, “Magnetization effects in electron cooling,” Fiz. Plazmy **4** (1978), p. 492; Sov. J. Plasma Phys. **4** (1978), 273.

I. Meshkov, “Electron Cooling; Status and Perspectives,” Phys. Part. Nucl. **25** (1994), 631.

Previous work: parametric model for including thermal effects

$$\mathbf{F} = -4Z^2 n_e m_e (r_e c^2)^2 \ln \left(\frac{\rho_{\max} + \rho_{\min} + r_L}{\rho_{\min} + r_L} \right) \frac{\mathbf{V}_{ion}}{(V_{ion}^2 + V_{eff}^2)^{3/2}}$$

$$\rho_{\min} = (Ze^2/4\pi\epsilon_0)/m_e V_{ion}^2$$

$$\rho_{\max} = V_{ion}/\max(\omega_{pe}, 1/\tau)$$

$$r_L = V_{rms,e,\perp}/\Omega_L(B_{\parallel})$$

$$V_{eff}^2 = V_{e,rms,\parallel}^2 + \Delta V_{\perp e}^2$$

V.V. Parkhomchuk, “New insights in the theory of electron cooling,” Nucl. Instr. Meth. in Phys. Res. **A 441** (2000).

I. Meshkov, “Electron Cooling; Status and Perspectives,” Phys. Part. Nucl. **25** (1994), 631.

Our approach is motivated by the work of *Ya. Derbenev*

THEORY OF ELECTRON COOLING

Ya. Derbenev, “Theory of Electron Cooling,” arXiv (2017);
<https://arxiv.org/abs/1703.09735>

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- The E-fields associated with friction must be carefully identified
 - *these are the fields generated by the presence of the ion*

bulk fields

friction

statistical fluctuations

$$\vec{E}(\vec{r}, \vec{v}, t) = \langle \vec{E}^0 \rangle(\vec{r}, t) + \langle \Delta \vec{E} \rangle(\vec{r}, \vec{v}, t) + \vec{E}^{fl}(\vec{r}, \vec{v}, t) \quad (1.1)$$

- Friction force must be calculated along the ion trajectory:

$$\vec{F} = -ze \langle \Delta \vec{E} \rangle(\vec{r}, \vec{v}, t) \Big|_{\vec{r}=\vec{r}(t), \vec{r}'(t)=\vec{v}} \quad (1.2)$$

- *we do this numerically for each individual ion-electron interaction*
 - **total force obtained by summing over e⁻ distribution (i.e. no shielding)**
- *bulk forces are removed by subtracting force from unperturbed e⁻'s*

Our model: strongly magnetized, relativistic cooling regime

→ short interaction time, strong magnetic field

- Prototyping is done in the parameter regime of Fedotov *et al.* (2006)

PHYSICAL REVIEW SPECIAL TOPICS - ACCELERATORS AND BEAMS 9, 074401 (2006)

Numerical study of the magnetized friction force

A. V. Fedotov,¹ D. L. Bruhwiler,² A. O. Sidorin,³ D. T. Abell,² I. Ben-Zvi,¹ R. Busby,² J. R. Cary,^{2,4} and V. N. Litvinenko¹

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Proceedings of HB2006, Tsukuba, Japan

Analysis of the magnetized friction force *

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A.O. Sidorin, JINR, Dubna, Russia

- For testing, we considered the following beam frame parameters:
 - *e^- density, $n_e = 2 \times 10^{15} \text{ m}^{-3}$*
 - *ideal solenoid, $B = 5 \text{ T}$*
 - *interaction time, $T_{\text{int}} = 4 \times 10^{-10} \text{ s} \sim 56 T_L \sim 0.16 T_{pl}$*
 - **16% of a plasma period → no shielding of the interaction**
 - *distance to nearest e^- , $r_l \sim 4.9 \times 10^{-6} \text{ m} \sim 10 r_L$*
 - **small Larmor radius → strong B-field assumption is reasonable**

Gyrokinetic averaging yields 1D e⁻ oscillations

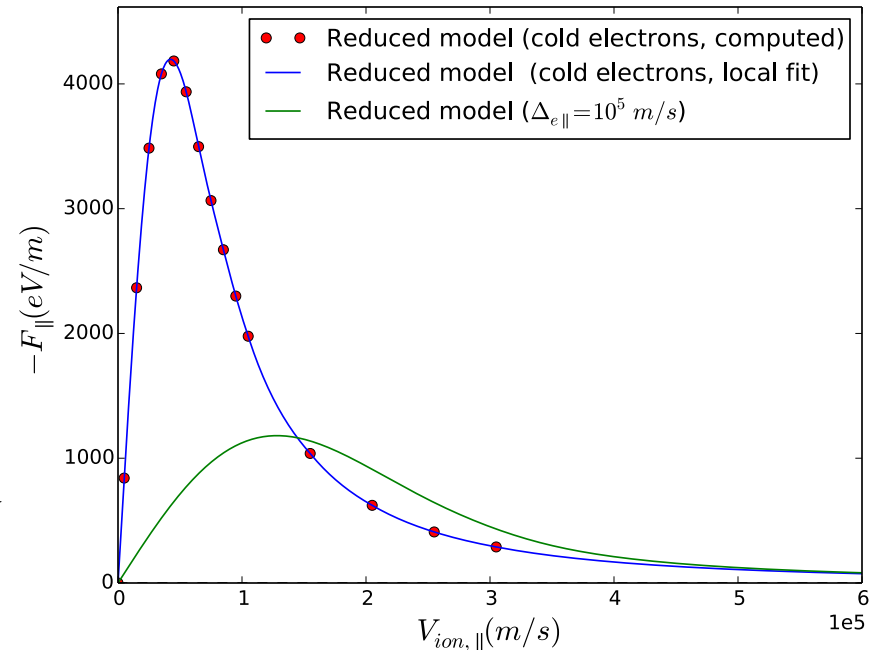
- Hamiltonian perturbation theory for single ion & e⁻
 - *unperturbed motion: drifting ion and magnetized e⁻*
 - *primary assumption: D (impact parameter) $\gg r_L$ (Larmor radius)*
 - *longitudinal dynamics: $V_{ion,\perp} = 0$ (to be relaxed in future work)*
- e⁻ gyrocenters stay on cylinder of constant radius D (different for different e⁻'s)
- choose ion to be stationary at the origin (convenient)
 - *gyrocenters move in a 1D potential:*
 - *weak nonlinearity (larger amplitudes => longer oscillation periods)*
- shortest oscillation period:
 - *both trapped and passing orbits*
 - *1D numerical simulations are required to capture these effects*
 - *no logarithmic divergence at either small or large impact parameters*

$$\ddot{z}(t) = -\frac{Ze^2}{4\pi\epsilon_0 m_e} \frac{z}{(D^2 + z^2)^{3/2}}$$

$$T_{lin} = \frac{2\pi}{c} \sqrt{\frac{D^3}{Zr_e}}$$

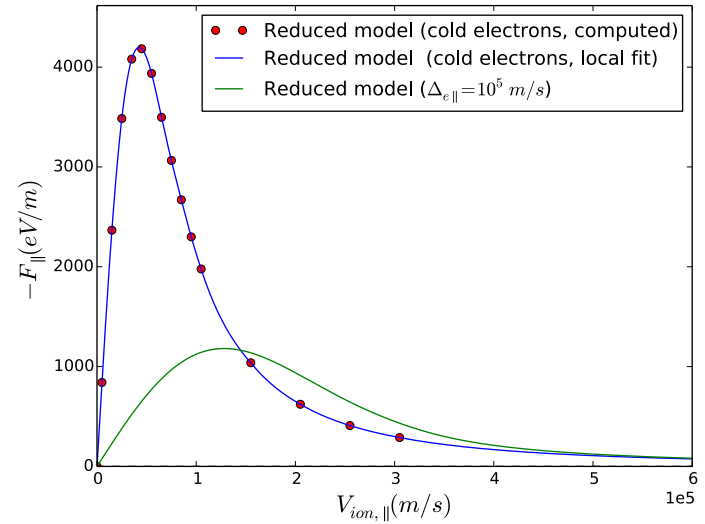
Correct behavior of $F_{\parallel}(V_{ion,\parallel})$ seen for both small and large $V_{ion,\parallel}$, cold and warm electrons

- For *cold* electrons and Au^{+79} ion, correct/expected qualitative behavior of $F_{\parallel}(V_{ion,\parallel})$ seen for both small and large $V_{ion,\parallel}$:
 - *linear in V for small V*
 - *$1/V^2$ for large V*
- For an arbitrary distribution $f(v_{e,\parallel})$ of *warm* electrons, $F_{\parallel}(V_{ion,\parallel})$ is computed by convolution of $f(v_{e,\parallel})$ with $F_{\parallel}(V_{ion,\parallel})$ for cold electrons
- Convolution with $f(v_{e,\parallel})$ acts as a smoothing filter $\Rightarrow$ peak of $F_{\parallel}(V_{ion,\parallel})$ for warm electrons is lower and shifted towards larger $V_{ion,\parallel}$
- Just as for cold e^- gas, for warm electrons $F_{\parallel}(V_{ion,\parallel})$ is linear in $V_{ion,\parallel}$ for small $V_{ion,\parallel}$ and scales as $1/V^2$ in the large $V_{ion,\parallel}$ region
- As expected, $F_{\parallel}(V_{ion,\parallel})$ for different electron temperatures converge as $V_{ion,\parallel}$ gets larger

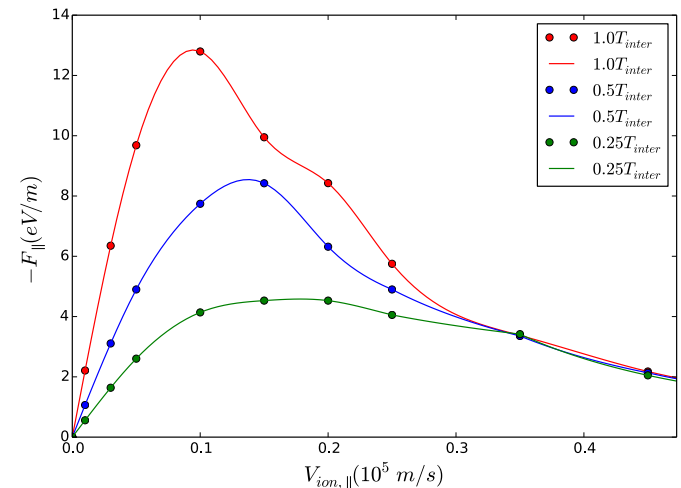


$F_{\parallel}(V_{ion,\parallel})$ for cold electrons: scaling in Z and T_{int}

- For cold electrons, looked at **protons** and **Au⁺⁷⁹** ion and different interaction times in the cooler (interaction-time-averaged force):
 - for small $V_{ion,\parallel}$: $F_{\parallel}(V) \sim V$; slope $dF_{\parallel}(V)/dV \approx 2Z n_e m_e r_e c^2 T_{int}$
 - large- V tail is well approximated by $F_{\parallel} \approx 2\pi Z^2 n_e m_e (r_e c^2)^2 / V^2$, with no dependence on T_{int}
 - for a given T_{int} , peak friction force scales approximately as $Z^{4/3}$
- For $T_{int} < T_{pl}$ and small-to-moderate V_{ion} , $F_{\parallel}(V_{ion,\parallel})$ goes up with interaction time; large- V tail is T_{int} -independent
- $F_{\parallel}(V_{ion,\parallel})$ is linear in n_e by construction



Above: Gold ion, cold and warm e-'s

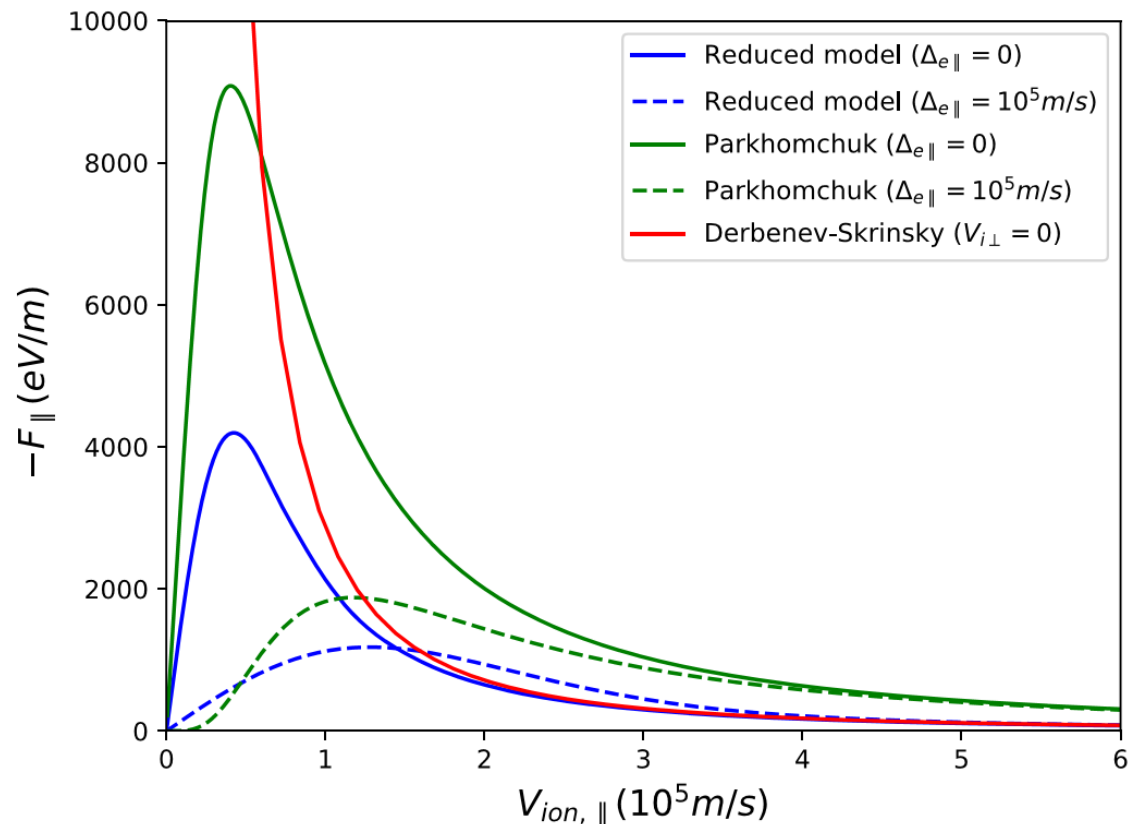


Above: Protons, cold e-'s, varying T_{int}

Compare with Derbenev-Skrinsky and Parkhomchuk (1)

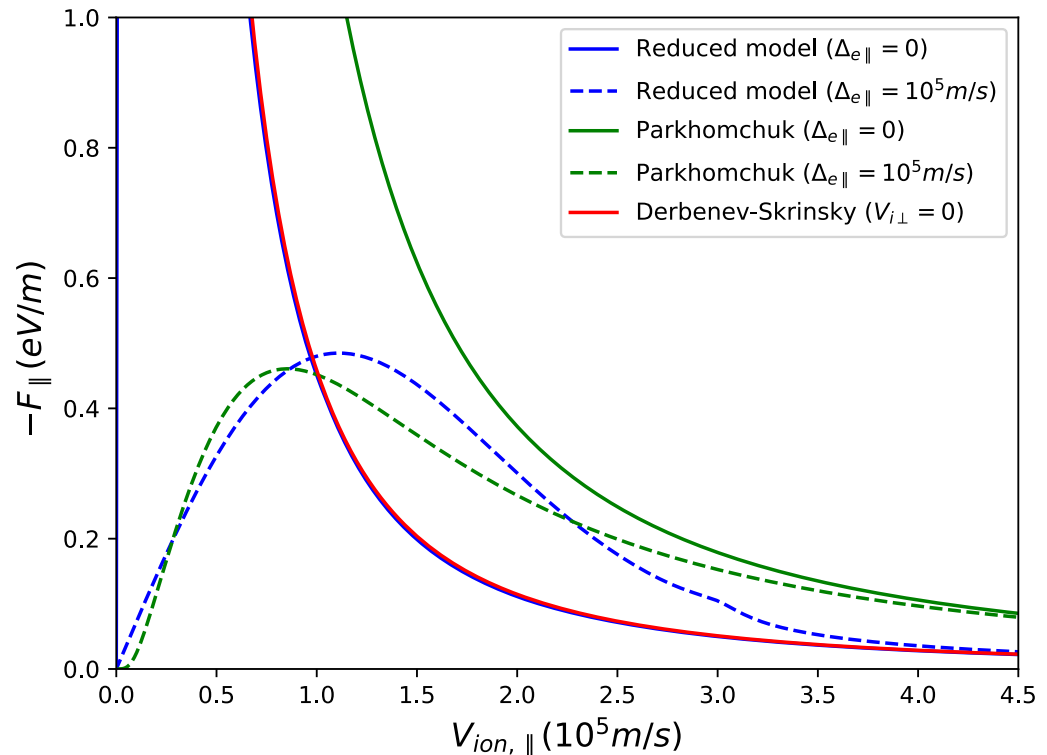
- Comparison of new model for an Au^{+79} ion, with:
 - Derbenev and Skrinsky (D&S) for $V_{ion,\perp} = 0$ and large $V_{ion,\parallel}$*
 - Parkhomchuk (P) with 0 and finite effective longitudinal e^- temperature*

- All models $\sim V^{-2}$ for large V
 - our simulation and semi-analytic model agree exactly with D&S*
 - consistently lower force than Parkhomchuk in this limit*
 - for lower ion velocity and warm electrons, details depend on Z*
- Parkhomchuk has unphysical inflection as $v \rightarrow 0$
 - can be corrected via constant Coulomb log*
 - no Coulomb log for our model*



Compare with Derbenev-Skrinsky and Parkhomchuk (2)

- Comparison of new model for **protons (zoomed in)**, with:
 - *Derbenev and Skrinsky (D-S)* for $V_{ion,\perp} = 0$ and large $V_{ion,\parallel}$
 - *Parkhomchuk (P)* with 0 and finite effective long. e^- temp
- In agreement with *D-S*, consistently lower force than *P* for larger $V_{ion,\parallel}$
- For warm electrons, new model agrees approximately with Parkhomchuk (but details depend on **Z**)
 - *not yet known how often (in what sub-domain of parameter space) this occurs*



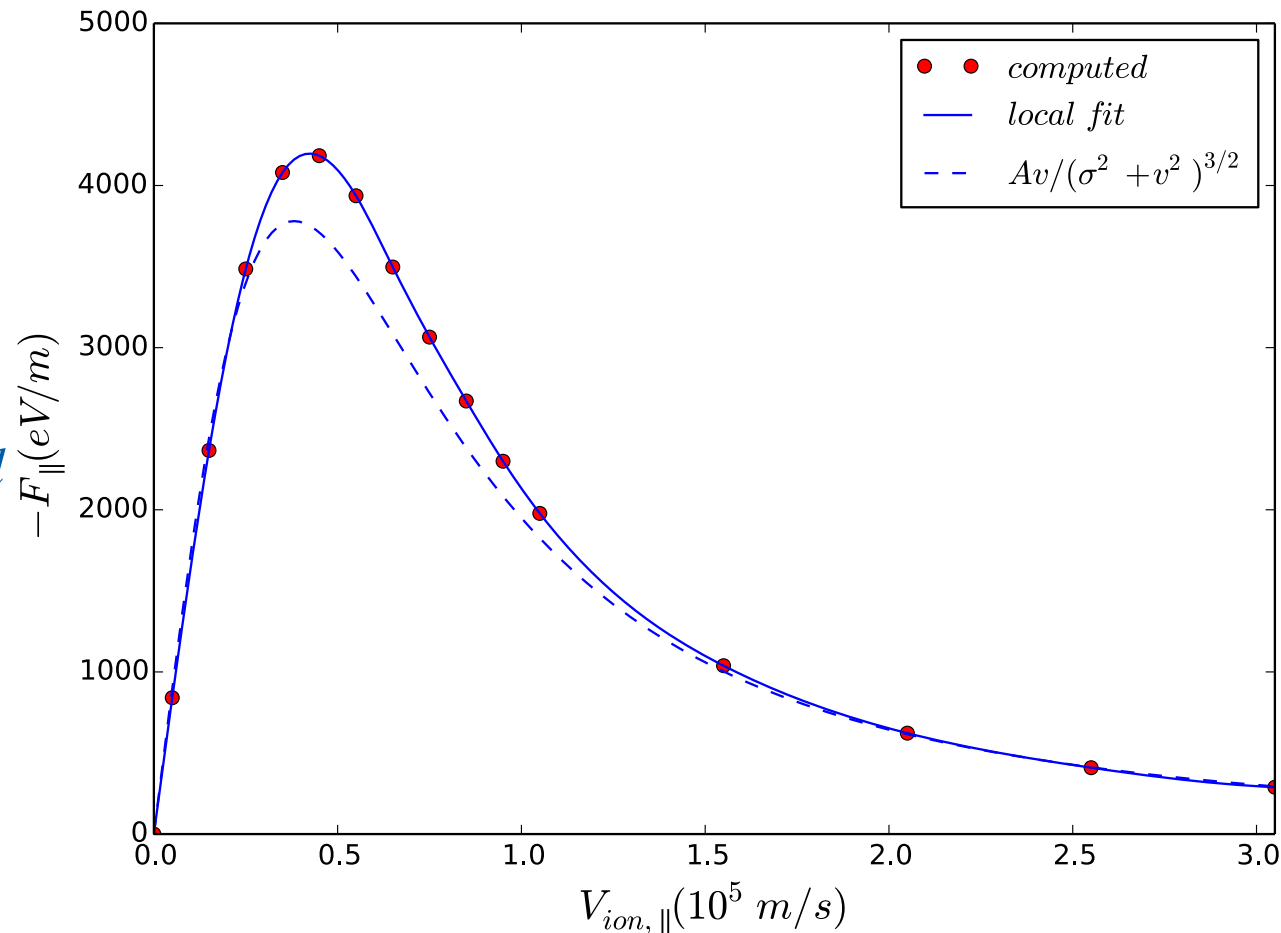
Simple, approximate 2-parameter model

$$F_{\parallel}(v) = \frac{Av}{(\sigma^2 + v^2)^{3/2}}$$

$$A = 2\pi Z^2 n_e m_e (r_e c^2)^2$$

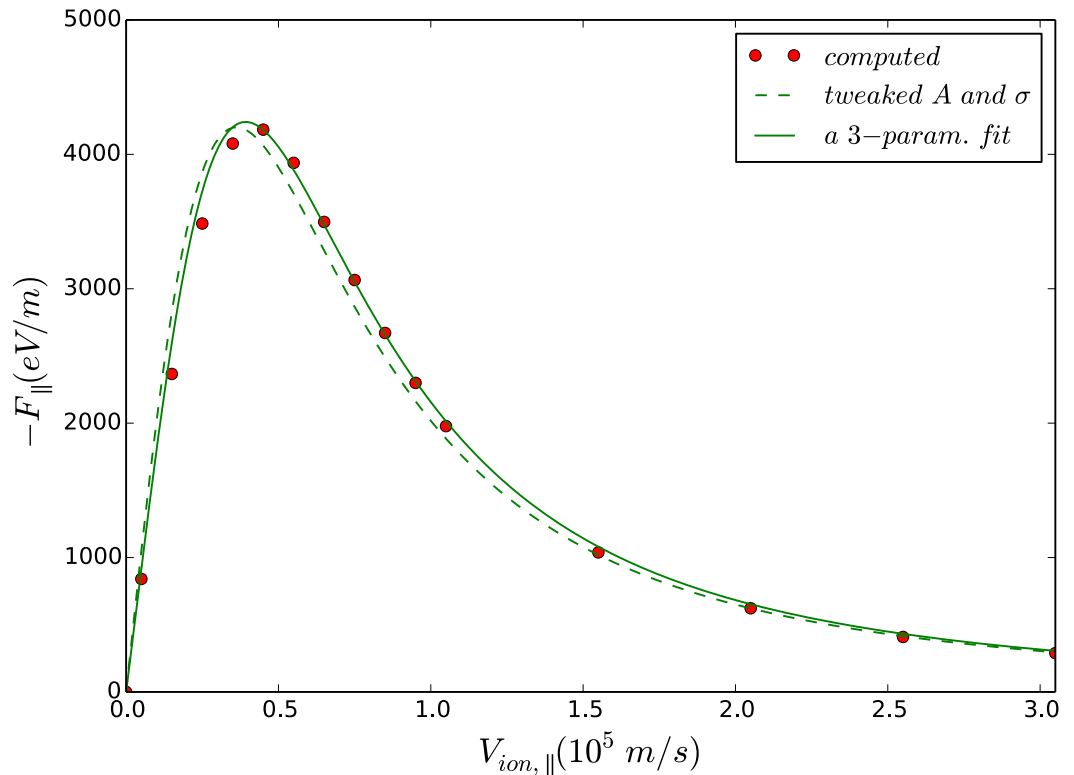
$$\sigma \approx (\pi Z r_e c^2 / T_{int})^{1/3}$$

- Large v :
 - $F_{\parallel} \sim A/v^2$
 - A found via fit and dimensions and scaling analysis
- Small v :
 - $dF/dv \sim A/\sigma^3$
 - σ found via fit and dimensions and scaling analysis
- Peak force is underestimated by ~10-15%



3-parameter model fits the calculations closely

- The physical system depends on 3 parameters:
 - n_e, Z, T_{int}
- Captured via perturbation of 2-parameter model:
 - *Tweak values of A and σ or add a small 3rd parameter*
- Improved parametric models are under development



Work in progress and future plans

- Improved parametrized models for cold electrons, and parametrized models for non-zero electron temperature
- Better understanding of the role of trapped (oscillatory) vs unbound electron orbits
- The case of finite B
- What happens to the magnitude of dynamic friction force as the interaction time approaches/exceeds T_{pl} ?
- Modeling transverse dynamic friction (have to work with non-zero electron temperature from the start)
- Statistical properties of $F(V)$: so far, only the expectation value was considered (in essence, the continuum limit)
- Adding new models to JSPEC as they become available, simulations in the EIC parameter regime
 - <https://sirepo.com>

Also a poster presentation this afternoon: TUPLO14

Thank You!

Questions?